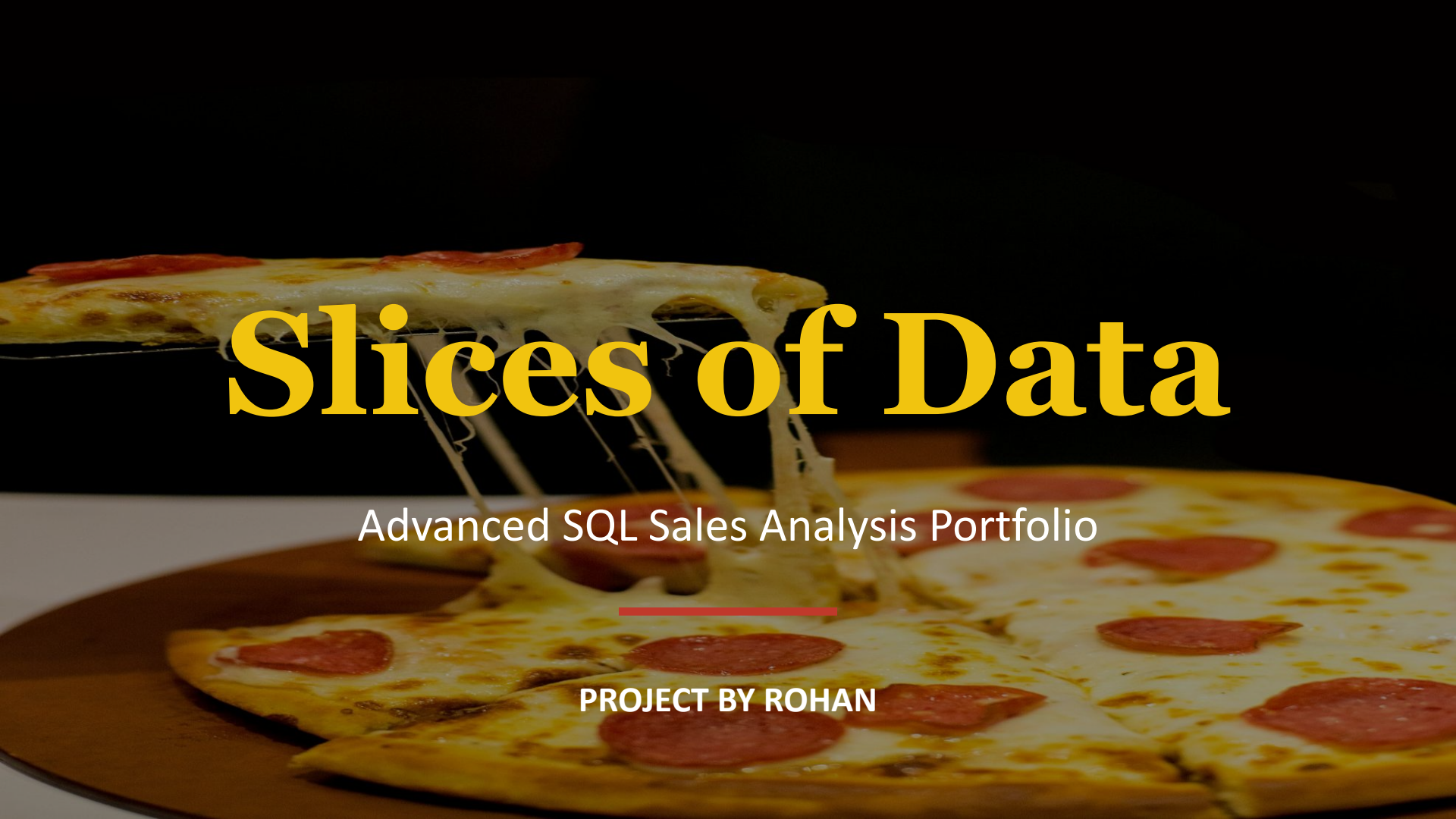

Slices of Data

Advanced SQL Sales Analysis Portfolio

PROJECT BY ROHAN

The Database Architecture

Analyzing 21,350 transaction records across 4 relational tables:

- Orders: Date & Time timestamps
- Details: Quantities per Pizza ID
- Pizzas: Size and Pricing matrix
- Types: Ingredients & Categorization

01. Total Order Volume

Business Question:

How many orders did we move over the year?

```
SELECT COUNT(order_id) AS TOTAL_SALES FROM orders;
```

The Result:

21,350 Orders

Human Insight:

This establishes our operational scale. It gives us a solid baseline for capacity planning and inventory management.

02. Annual Gross Revenue

Business Question:

What is the total revenue generated from all sales?

```
SELECT ROUND(SUM(o.quantity * p.price)) AS total_revenue
FROM orders_details o JOIN pizzas p ON o.pizza_id = p.pizza_id;
```

The Result:

$817,860

Human Insight:

Total annual revenue crossed the $800k mark. This represents the primary indicator of store health and financial success.

03. The Premium Choice

Business Question:

Which pizza carries the highest price tag?

```
SELECT name, MAX(price) FROM pizza_types JOIN pizzas
GROUP BY name ORDER BY max_price DESC LIMIT 1;
```

The Result:

The Greek Pizza ($35.95)

Human Insight:

The XXL Greek pizza represents our ceiling for customer spending. Knowing our luxury price points helps in high-end positioning.

04. Sizing Up Demand

Business Question:

Which pizza size (S, M, L) do our customers prefer?

```
SELECT size, SUM(quantity) FROM details JOIN pizzas
GROUP BY size ORDER BY total_qty DESC;
```

The Result:

Large (L): 18,956 Units Sold

Human Insight:

Customers clearly prefer Large pizzas. This informs our procurement to prioritize 14" boxes and larger dough portions.

05. Best-Sellers (Top 5)

Business Question:

Which five pizza types are ordered most frequently?

```
SELECT name, SUM(quantity) FROM pizza_types JOIN orders_details
GROUP BY name ORDER BY SUM(quantity) DESC LIMIT 5;
```

The Result:

1. Classic Deluxe (2,453 Sold)

Human Insight:

These 'Hero' products drive our daily volume. The Classic Deluxe is the clear menu champion.

06. Segment Performance

Business Question:

How many pizzas were sold in each category?

```
SELECT category, SUM(quantity) FROM pizza_types JOIN details
GROUP BY category ORDER BY SUM(quantity) DESC;
```

The Result:

Classic: 14,888 | Supreme: 11,987

Human Insight:

The Classic category dominates volume, proving that traditional recipes hold the broadest demographic appeal.

07. Peak Service Hours

Business Question:

At what hours does the store see the most activity?

```
SELECT HOUR(order_time), COUNT(order_id) FROM orders
GROUP BY HOUR(order_time);
```

The Result:

Peaks: 12:00 PM and 6:00 PM

Human Insight:

Staffing should be maximized for Lunch and Dinner. We see a significant lull at 3 PM for staff breaks.

08. Menu Diversity

Business Question:

How many unique recipes exist per category?

```
SELECT category, COUNT(name) FROM pizza_types
GROUP BY category;
```

The Result:

Supreme: 9 | Classic: 8 | Veggie: 8

Human Insight:

We maintain a balanced portfolio. Supreme offers the most variety, ensuring customers always have something new to try.

09. Daily Throughput

Business Question:

What is the average number of pizzas prepared per day?

```
SELECT AVG(daily_total) FROM (SELECT date, SUM(qty) GROUP BY date);
```

The Result:

138 Pizzas / Day

Human Insight:

Kitchen readiness for ~140 units daily optimizes ingredient fermentation and minimizes waste.

10. The Revenue Powerhouses

Business Question:

Which three pizzas bring in the most cash?

```
SELECT name, SUM(price * qty) AS revenue FROM ...
GROUP BY name ORDER BY revenue DESC LIMIT 3;
```

The Result:

Thai Chicken ($43,434), BBQ Ckn, Cali Ckn

Human Insight:
Chicken pizzas are our financial anchors. Their premium price offsets volume to lead the revenue charts.

11. Market Contribution

Business Question:

What % of revenue does each category contribute?

```
SELECT category, (SUM(sales)/TOTAL) * 100 FROM ...
```

The Result:

Classic: 26.9% | Supreme: 25.5%

Human Insight:

Our business is remarkably balanced. Every category contributes roughly a quarter of the annual $817k revenue.

12. Growth Scaling

Business Question:

How does our revenue build over the year?

```
SELECT order_date, SUM(revenue) OVER (ORDER BY order_date);
```

The Result:

Final Total: $817,860

Human Insight:
Steady linear growth suggests a highly stable customer base with zero major seasonal slow-downs.

13. Category Champions

Business Question:

Which three pizzas are highest earners per category?

```
SELECT category, name, RANK() OVER(PARTITION BY category...)
```

The Result:

Stars: Classic Deluxe, Thai Chicken

Human Insight:

This ranking isolates the 'Anchors' of each segment, critical for kitchen focus and promotions.

Strategic Takeaways

- Staffing: Optimize for 12 PM & 6 PM peaks.
- Revenue: Focus on Chicken Pizzas ($40k+ earners).
- Inventory: Priority stock for Large (L) size boxes.
- Growth: Stable $817k/year with healthy menu variety.

Let's Connect!

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